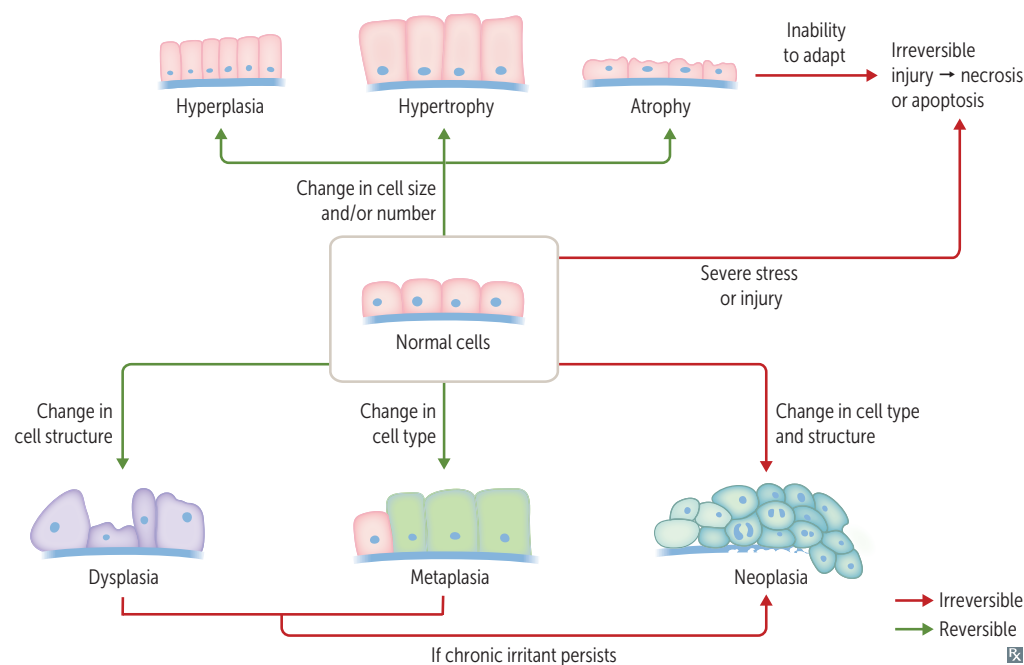


## ► PATHOLOGY—CELLULAR INJURY

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Cellular adaptations</b> | Reversible changes that can be physiologic (eg, uterine enlargement during pregnancy) or pathologic (eg, myocardial hypertrophy 2° to systemic HTN). If stress is excessive or persistent, adaptations can progress to cell injury (eg, significant LV hypertrophy → myocardial injury → HF).                                                                                                                                                                                                                                                                                                                                     |
| <b>Hypertrophy</b>          | ↑ structural proteins and organelles → ↑ in size of cells. Example: cardiac hypertrophy.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Hyperplasia</b>          | Controlled proliferation of stem cells and differentiated cells → ↑ in number of cells (eg, benign prostatic hyperplasia). Excessive stimulation → pathologic hyperplasia (eg, endometrial hyperplasia), which may progress to dysplasia and cancer.                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Atrophy</b>              | ↓ in tissue mass due to ↓ in size (↑ cytoskeleton degradation via ubiquitin-proteasome pathway and autophagy [programmed digestion of damaged/misfolded intracellular proteins]; ↓ protein synthesis) and/or number of cells (apoptosis). Causes include disuse, denervation, loss of blood supply, loss of hormonal stimulation, poor nutrition.                                                                                                                                                                                                                                                                                 |
| <b>Metaplasia</b>           | Reprogramming of stem cells → replacement of one cell type by another that can adapt to a new stressor. Usually due to exposure to an irritant, such as gastric acid (→ esophageal epithelium replaced by intestinal epithelium, called Barrett esophagus) or tobacco smoke (→ respiratory ciliated columnar epithelium replaced by stratified squamous epithelium). May progress to dysplasia → malignant transformation with persistent insult (eg, Barrett esophagus → esophageal adenocarcinoma). Metaplasia of connective tissue can also occur (eg, myositis ossificans, the formation of bone within muscle after trauma). |
| <b>Dysplasia</b>            | Disordered, precancerous epithelial cell growth; not considered a true adaptive response. Characterized by loss of uniformity of cell size and shape (pleomorphism); loss of tissue orientation; nuclear changes (eg, ↑ nuclear:cytoplasmic ratio and clumped chromatin). Mild and moderate dysplasias (ie, do not involve entire thickness of epithelium) may regress with alleviation of inciting cause. Severe dysplasia often becomes irreversible and progresses to carcinoma in situ. Usually preceded by persistent metaplasia or pathologic hyperplasia.                                                                  |

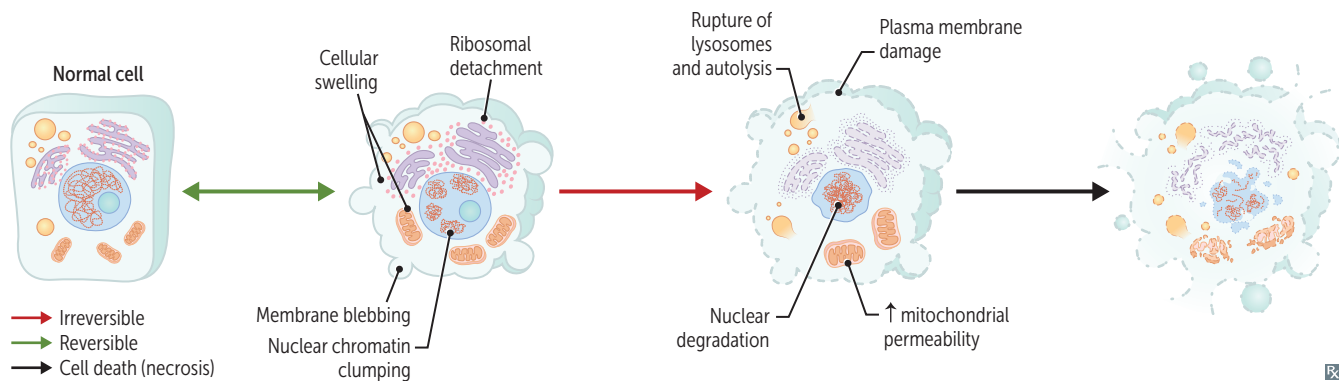


**Cell injury****Reversible cell injury**

- $\downarrow$  ATP  $\rightarrow$   $\downarrow$  activity of  $\text{Ca}^{2+}$  and  $\text{Na}^+/\text{K}^+$ -ATPase pumps  $\rightarrow$  cellular swelling (cytosol, mitochondria, endoplasmic reticulum/Golgi), which is the earliest morphologic manifestation
- Ribosomal/polysomal detachment  $\rightarrow$   $\downarrow$  protein synthesis
- Plasma membrane changes (eg, blebbing)
- Nuclear changes (eg, chromatin clumping)
- Rapid loss of function (eg, myocardial cells are noncontractile after 1–2 minutes of ischemia)
- Myelin figures (aggregation of peroxidized lipids)

**Irreversible cell injury**

- Breakdown of plasma membrane  $\rightarrow$  cytosolic enzymes (eg, troponin) leak outside of cell, influx of  $\text{Ca}^{2+}$   $\rightarrow$  activation of degradative enzymes
- Mitochondrial damage/dysfunction  $\rightarrow$  loss of electron transport chain  $\rightarrow$   $\downarrow$  ATP
- Rupture of lysosomes  $\rightarrow$  autolysis
- Nuclear degradation: pyknosis (nuclear condensation)  $\rightarrow$  karyorrhexis (nuclear fragmentation caused by endonuclease-mediated cleavage)  $\rightarrow$  karyolysis (nuclear dissolution)
- Amorphous densities/inclusions in mitochondria



## Apoptosis

ATP-dependent programmed cell death.

Intrinsic, extrinsic, and perforin/granzyme B pathways → activate caspases (cytosolic proteases)  
→ cellular breakdown including cell shrinkage, chromatin condensation, membrane blebbing, and formation of apoptotic bodies, which are then phagocytosed.

Characterized by deeply eosinophilic cytoplasm and basophilic nucleus, pyknosis, and karyorrhexis. Cell membrane typically remains intact without significant inflammation (unlike necrosis).

DNA laddering (fragments in multiples of 180 bp) is a sensitive indicator of apoptosis.

### Intrinsic (mitochondrial) pathway

Involved in tissue remodeling in embryogenesis. Occurs when a regulating factor is withdrawn from a proliferating cell population (eg, ↓ IL-2 after a completed immunologic reaction → apoptosis of proliferating effector cells). Also occurs after exposure to injurious stimuli (eg, radiation, toxins, hypoxia).

Regulated by Bcl-2 family of proteins. **Bax** and **Bak** are proapoptotic (**Ba**d for survival), while **Bcl-2** and **Bcl-xL** are antiapoptotic (**Be** clever, **live**).

BAX and BAK form pores in the mitochondrial membrane → release of cytochrome C from inner mitochondrial membrane into the cytoplasm → activation of caspases.

Bcl-2 keeps the mitochondrial membrane impermeable, thereby preventing cytochrome C release.

Bcl-2 overexpression (eg, follicular lymphoma t[14;18]) → ↓ caspase activation → tumorigenesis.

### Extrinsic (death receptor) pathway

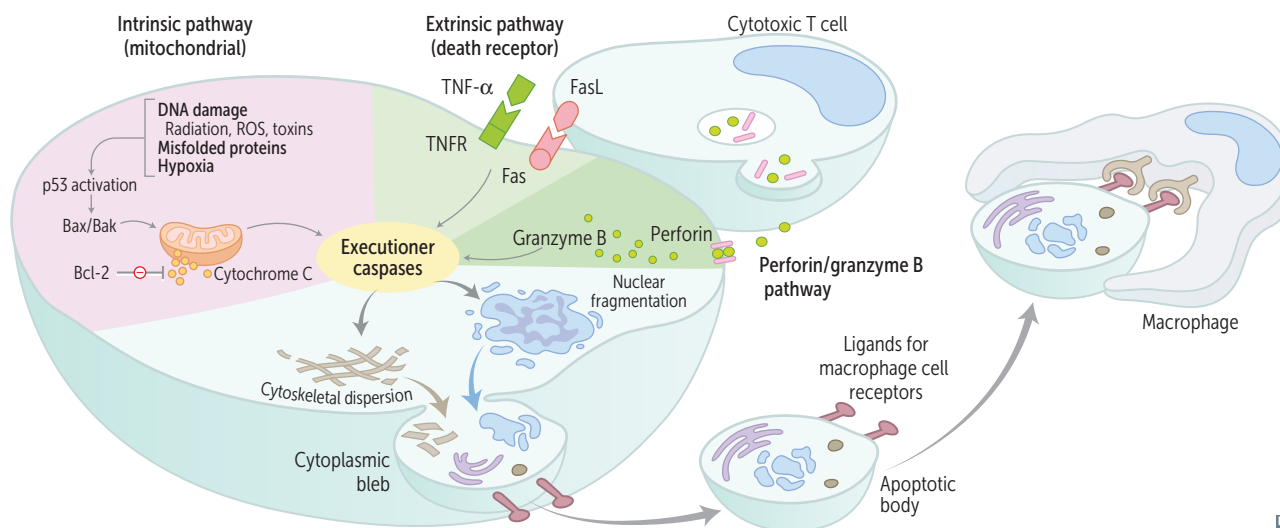
Ligand receptor interactions: FasL binding to Fas (CD95) or TNF-α binding to its receptor.

Fas-FasL interaction is necessary in thymic medullary negative selection.

**Autoimmune lymphoproliferative syndrome**—caused by defective Fas-FasL interaction → failure of clonal deletion → ↑ numbers of self-reacting lymphocytes. Presents with lymphadenopathy, hepatosplenomegaly, autoimmune cytopenias.

### Perforin/granzyme B pathway

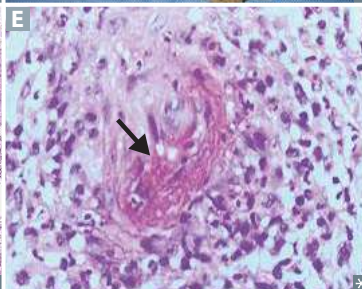
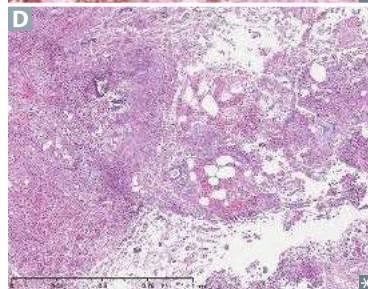
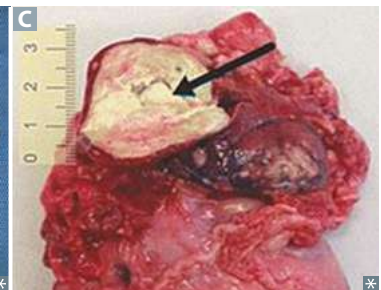
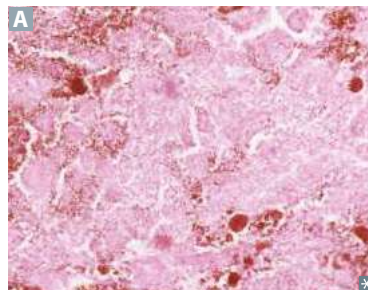
Release of granules containing perforin and granzyme B by immune cells (cytotoxic T-cell and natural killer cell) → perforin forms a pore for granzyme B to enter the target cell.

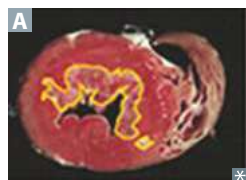


**Necrosis**

Exogenous injury → plasma membrane damage → intracellular components leak → cell undergoes enzymatic degradation and protein denaturation → local inflammatory reaction (unlike apoptosis).

| TYPE                | SEEN IN                                                                                                                         | DUE TO                                                                                                                                                  | HISTOLOGY                                                                                                                                                    |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Coagulative</b>  | Ischemia/infarcts in most tissues (except brain)                                                                                | Ischemia or infarction; injury denatures enzymes → proteolysis blocked                                                                                  | Preserved cellular architecture (cell outlines seen), but nuclei disappear; ↑ cytoplasmic binding of eosin stain (→ ↑ eosinophilia; red/pink color) <b>A</b> |
| <b>Liquefactive</b> | Bacterial abscesses, CNS infarcts                                                                                               | Neutrophils release lysosomal enzymes that digest the tissue                                                                                            | Early: cellular debris and macrophages<br>Late: cystic spaces and cavitation (CNS) <b>B</b><br>Neutrophils and cell debris seen with bacterial infection     |
| <b>Caseous</b>      | TB, systemic fungi (eg, <i>Histoplasma capsulatum</i> ), <i>Nocardia</i>                                                        | Macrophages wall off the infecting microorganism → granular debris                                                                                      | Fragmented cells and debris surrounded by lymphocytes and macrophages (granuloma)<br>Cheeselike gross appearance <b>C</b>                                    |
| <b>Fat</b>          | Enzymatic: acute pancreatitis (saponification of peripancreatic fat)<br>Nonenzymatic: traumatic (eg, injury to breast tissue)   | Damaged pancreatic cells release lipase, which breaks down triglycerides; liberated fatty acids bind calcium → saponification (chalky-white appearance) | Outlines of dead fat cells without peripheral nuclei; saponification of fat (combined with $\text{Ca}^{2+}$ ) appears dark blue on H&E stain <b>D</b>        |
| <b>Fibrinoid</b>    | Immune vascular reactions (eg, polyarteritis nodosa)<br>Nonimmune vascular reactions (eg, hypertensive emergency, preeclampsia) | Immune complex deposition (type III hypersensitivity reaction) and/or plasma protein (eg, fibrin) leakage from damaged vessel                           | Vessel walls contain eosinophilic layer of proteinaceous material <b>E</b>                                                                                   |
| <b>Gangrenous</b>   | Distal extremity and GI tract, after chronic ischemia                                                                           | Dry: ischemia <b>F</b><br>Wet: superinfection                                                                                                           | Coagulative<br>Liquefactive superimposed on coagulative                                                                                                      |



**Ischemia**

Inadequate blood supply to meet demand. Mechanisms include ↓ arterial perfusion (eg, atherosclerosis), ↓ venous drainage (eg, testicular torsion, Budd-Chiari syndrome), shock. Regions most vulnerable to hypoxia/ischemia and subsequent infarction:

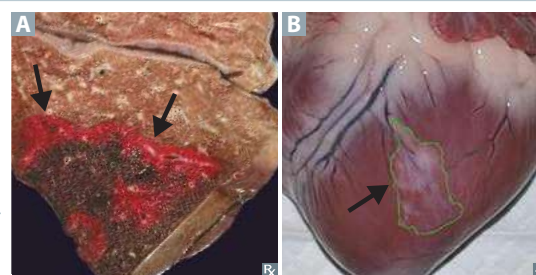
| ORGAN  | SUSCEPTIBLE REGION                                                                               |
|--------|--------------------------------------------------------------------------------------------------|
| Brain  | ACA/MCA/PCA boundary areas <sup>a,b</sup>                                                        |
| Heart  | Subendocardium of LV (yellow lines in <b>A</b> outline a subendocardial infarction)              |
| Kidney | Straight segment of proximal tubule (medulla)<br>Thick ascending limb (medulla)                  |
| Liver  | Area around central vein (zone III)                                                              |
| Colon  | Splenic flexure (Griffith point), <sup>a</sup> rectosigmoid junction (Sudeck point) <sup>a</sup> |

<sup>a</sup>Watershed areas (border zones) receive blood supply from most distal branches of 2 arteries with limited collateral vascularity. These areas are susceptible to ischemia from hypoperfusion.

<sup>b</sup>Neurons most vulnerable to hypoxic-ischemic insults include Purkinje cells of the cerebellum and pyramidal cells of the hippocampus and neocortex (layers 3, 5, 6).

**Types of infarcts****Red infarct**

Occurs in venous occlusion and tissues with multiple blood supplies (eg, liver, lung **A**, intestine, testes), and with reperfusion (eg, after angioplasty). **R**eperfusion injury is due to damage by free radicals.

**Pale infarct**

Occurs in solid organs with a single (end-arterial) blood supply (eg, heart **B**, kidney).

**Free radical injury**

Free radicals damage cells via membrane lipid peroxidation, protein modification, DNA breakage. Initiated via radiation exposure (eg, cancer therapy), metabolism of drugs (phase I), redox reactions, nitric oxide (eg, inflammation), transition metals (eg, iron, copper; form free radicals via Fenton reaction), WBC (eg, neutrophils, macrophages) oxidative burst.

Free radicals can be eliminated by scavenging enzymes (eg, catalase, superoxide dismutase, glutathione peroxidase), spontaneous decay, antioxidants (eg, vitamins A, C, E), and certain metal carrier proteins (eg, transferrin, ceruloplasmin).

Examples:

- Oxygen toxicity: retinopathy of prematurity (abnormal vascularization), bronchopulmonary dysplasia, reperfusion injury after thrombolytic therapy
- Drug/chemical toxicity: acetaminophen overdose (hepatotoxicity), carbon tetrachloride (converted by cytochrome P-450 into CCl<sub>3</sub> free radical → fatty liver [cell injury → ↓ apolipoprotein synthesis → fatty change], centrilobular necrosis)
- Metal storage diseases: hemochromatosis (iron) and Wilson disease (copper)



Ionizing radiation toxicity

Ionizing radiation causes DNA (eg, double strand breaks) and cellular damage both directly and indirectly through the production of free radicals. Complications usually arise when patient is exposed to significant doses (eg, radiotherapy, nuclear reactor accidents):

- Localized inflammation and fibrosis
- Neoplasia (eg, leukemia, thyroid cancer)

**Acute radiation syndrome**—develops after sudden whole-body exposure to high doses of ionizing radiation → nausea, vomiting, diarrhea, hair loss, erythema, cytopenias, headache, altered mental status.

Stem cells of rapidly regenerating tissues (eg, skin, bone marrow, GI tract, gonads) are the most susceptible to radiation injury. Radiotherapy damages cancer cells more than healthy cells because cancer cells have dysfunctional DNA repair mechanisms in addition to high replicative rates.

Types of calcification

Calcium deposits appear deeply basophilic (arrow in **A**) on H&E stain.

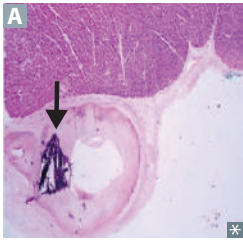
Dystrophic calcification

Metastatic calcification

Ca<sup>2+</sup> DEPOSITION

EXTENT

ASSOCIATED CONDITIONS



In abnormal (diseased) tissues

Tends to be localized (eg, calcific aortic stenosis)

TB (lung and pericardium) and other granulomatous infections, liquefactive necrosis of chronic abscesses, fat necrosis, infarcts, thrombi, schistosomiasis, congenital CMV, toxoplasmosis, rubella, psammoma bodies, CREST syndrome, atherosclerotic plaques can become calcified

In normal tissues

Widespread (ie, diffuse, metastatic)

Predominantly in interstitial tissues of kidney, lung, and gastric mucosa (these tissues lose acid quickly; ↑ pH favors Ca<sup>2+</sup> deposition)

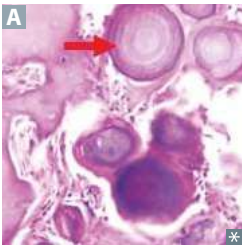
Nephrocalcinosis of collecting ducts may lead to nephrogenic diabetes insipidus and renal failure

ETIOLOGY

2° to injury or necrosis

2° to hyperphosphatemia (eg, chronic kidney disease) or hypercalcemia (eg, 1° hyperparathyroidism, sarcoidosis, hypervitaminosis D)

Psammoma bodies



Concentrically laminated calcified spherules **A**.

Usually seen in certain types of tumors:

- Papillary carcinoma of thyroid gland and kidney
- Somatostatinoma
- Meningioma
- Mesothelioma
- Ovarian serous papillary cystadenocarcinoma
- Milk (prolactinoma)

Histology: Basophilic staining of Ca<sup>2+</sup> deposits within tissue.

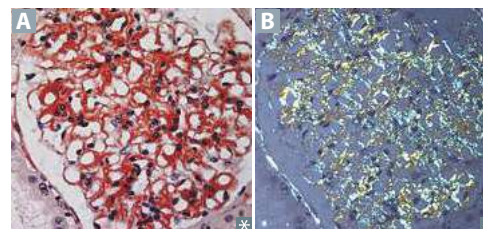
PSoMMOMa Bodies.

**Amyloidosis**

Extracellular deposition of protein in abnormal fibrillar form ( $\beta$ -pleated sheet configuration) → cell injury and apoptosis. Manifestations vary depending on involved organ and include:

- Renal—nephrotic syndrome.
- Cardiac—restrictive cardiomyopathy.
- GI—hepatosplenomegaly.
- Neurologic—peripheral neuropathy.
- Musculoskeletal—muscle enlargement (eg, macroglossia), carpal tunnel syndrome.
- Skin—waxy thickening, easy bruising.

Amyloid deposits are visualized by Congo red stain (red/orange on nonpolarized light **A**, apple-green birefringence on polarized light **B**), and H&E stain (amorphous pink).



| COMMON TYPES                        | FIBRIL PROTEIN                              | NOTES                                                                                                                                                                                          |
|-------------------------------------|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Systemic</b>                     |                                             |                                                                                                                                                                                                |
| <b>Primary amyloidosis</b>          | <b>AL</b> (from Ig <b>L</b> ight chains)    | Seen in plasma cell dyscrasias (eg, multiple myeloma)                                                                                                                                          |
| <b>Secondary amyloidosis</b>        | <b>AA</b> (serum <b>A</b> myloid <b>A</b> ) | Seen in chronic inflammatory conditions, (eg, rheumatoid arthritis, IBD, familial Mediterranean fever, protracted infection)                                                                   |
| <b>Transthyretin amyloidosis</b>    | Transthyretin                               | Sporadic (wild-type <i>TTR</i> )—slowly progressive, associated with aging; mainly affects the heart<br>Hereditary (mutated <i>TTR</i> )—familial amyloid polyneuropathy and/or cardiomyopathy |
| <b>Dialysis-related amyloidosis</b> | $\beta_2$ -microglobulin                    | Seen in patients with ESRD on long-term dialysis                                                                                                                                               |
| <b>Localized</b>                    |                                             |                                                                                                                                                                                                |
| <b>Alzheimer disease</b>            | $\beta$ -amyloid protein                    | Cleaved from amyloid precursor protein                                                                                                                                                         |
| <b>Isolated atrial amyloidosis</b>  | ANP                                         | Common, associated with aging; ↑ risk for atrial fibrillation                                                                                                                                  |
| <b>Type 2 diabetes mellitus</b>     | Islet amyloid polypeptide                   | Caused by deposition of amylin in pancreatic islets                                                                                                                                            |
| <b>Medullary thyroid cancer</b>     | Calcitonin                                  | Secreted from tumor cells                                                                                                                                                                      |